Multivariate distributions

September 2026

CASE 1 — Poisson–Binomial Hierarchical Model (Conversions per Hour)

Model

$$X \sim \text{Poisson}(\lambda)$$

represent the number of users that visit an e-commerce website in an hour, i.e.,

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

Let Y denote the number of sales made in an hour. Given $X = x$ user visits, the conditional distribution of Y is given by

$$Y \mid X = x \sim \text{Binomial}(x, p),$$

i.e.,

$$P(Y = y \mid X = x) = \binom{x}{y} p^y (1-p)^{x-y}$$

1. Joint Distribution

$$P(X = x, Y = y) = \binom{x}{y} p^y (1-p)^{x-y} \frac{e^{-\lambda}\lambda^x}{x!}, \qquad 0 \le y \le x$$

2. Marginal Distribution of Y

We compute:

$$P(Y = y) = \sum_{x=y}^{\infty} P(X = x, Y = y)$$

Substitute:

$$= \sum_{x=y}^{\infty} \frac{x!}{y!(x-y)!} p^y (1-p)^{x-y} \frac{e^{-\lambda}\lambda^x}{x!}$$

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Cancel $x!$:

$$= \sum_{x=y}^{\infty} \frac{1}{y!(x-y)!} p^y (1-p)^{x-y} e^{-\lambda} \lambda^x$$

Let $k = x - y$:

$$= e^{-\lambda} \frac{(\lambda p)^y}{y!} \sum_{k=0}^{\infty} \frac{(\lambda(1-p))^k}{k!}$$

Recognize exponential series:

$$= e^{-\lambda} \frac{(\lambda p)^y}{y!} e^{\lambda(1-p)} = \frac{e^{-\lambda p} (\lambda p)^y}{y!}$$

Thus,

$$Y \sim \text{Poisson}(\lambda p)$$

Numerical Example

Let:

$$\lambda = 120, \qquad p = 0.05$$

Then:

$$Y \sim \text{Poisson}(6)$$

Probability of No Sales

General:

$$P(Y = 0) = e^{-\lambda p}$$

Numeric:

$$P(Y = 0) = e^{-6} = 0.00248$$

Probability of At Least 10 Sales

General:

$$P(Y \geq 10) = 1 - \sum_{y=0}^{9} \frac{e^{-\lambda p} (\lambda p)^y}{y!}$$

Numeric:

$$1 - \sum_{y=0}^{9} \frac{e^{-6} 6^y}{y!} \approx 0.083$$

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3. Conditional Distribution $X \mid Y = y$

By Bayes:

$$P(X = x \mid Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

Substitute and simplify:

$$P(X = x \mid Y = y) = \frac{e^{-\lambda(1-p)}(\lambda(1-p))^{x-y}}{(x-y)!}, \qquad x \geq y$$

Thus:

$$X \mid Y = y = y + \text{Poisson}(\lambda(1-p))$$

Numerical Example

$$\lambda(1-p) = 120(0.95) = 114$$

If 8 purchases observed:

$$X \mid Y = 8 = 8 + \text{Poisson}(114)$$

Probability total visitors = 125:

$$P(X = 125 \mid Y = 8) = \frac{e^{-114}114^{117}}{117!}$$

Interpretation: probability traffic was exactly 125 given 8 conversions.

CASE 2 — Casual vs Intent-Driven Browser (Exit Model)

$$X = \text{Number of Pages Viewed Before Exit}$$

1. Model Specification

A user visiting a website belongs to one of two latent types:

$$S \in \{C, I\}$$

which represent casual and intent-driven users respectively. Say we have a prior belief on the user type, perhaps based on historical data, given by

$$P(S = C) = \pi, \qquad P(S = I) = 1 - \pi$$

Each page view is followed by a decision:

- User exits with probability q

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- User continues browsing with probability $1 - q$

Exit probabilities differ by type:

$$\text{Casual user exit probability: } q_C$$

$$\text{Intent-driven user exit probability: } q_I$$

with

$$q_C > q_I$$

So casual users are more likely to leave early.
Define:

$$X = \text{number of pages viewed before exit}$$

2. Conditional Distributions

Given user type, the number of pages before exit is geometric:

$$X \mid S = C \sim \text{Geometric}(q_C)$$

$$X \mid S = I \sim \text{Geometric}(q_I)$$

Supported on $\{1, 2, 3, \dots\}$.
Thus:

$$P(X = k \mid S = C) = (1 - q_C)^{k-1} q_C$$

$$P(X = k \mid S = I) = (1 - q_I)^{k-1} q_I$$

3. Marginal Distribution (Mixture)

$$P(X = k) = \pi(1 - q_C)^{k-1} q_C + (1 - \pi)(1 - q_I)^{k-1} q_I$$

This is a mixture of two geometric distributions.
It is not geometric.

Numerical Specification

$$\pi = 0.6, \quad q_C = 0.6, \quad q_I = 0.2$$

- 60% users are casual
- Casual users exit quickly
- Intent users stay longer

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4. Probability User Exits After First Page

General:

$$P(X = 1) = \pi q_C + (1 - \pi)q_I$$

Substitute:

$$= 0.6(0.6) + 0.4(0.2)$$
$$= 0.36 + 0.08$$
$$= 0.44$$

So 44% leave immediately.

5. Probability User Views Exactly 3 Pages

General:

$$P(X = 3) = \pi(1 - q_C)^2 q_C + (1 - \pi)(1 - q_I)^2 q_I$$

Substitute:
Casual part:

$$(0.4)^2(0.6) = 0.16(0.6) = 0.096$$

Multiply by 0.6:

$$0.0576$$

Intent part:

$$(0.8)^2(0.2) = 0.64(0.2) = 0.128$$

Multiply by 0.4:

$$0.0512$$

Total:

$$0.1088$$

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6. Survival Probability

$$P(X > k) = \pi(1 - q_C)^k + (1 - \pi)(1 - q_I)^k$$

Example:

$$P(X > 3) = 0.6(0.4)^3 + 0.4(0.8)^3$$

$$= 0.6(0.064) + 0.4(0.512)$$

$$= 0.0384 + 0.2048$$

$$= 0.2432$$

7. Posterior Probability of Intent Given Long Session

Compute:

$$P(S = I \mid X = 5)$$

Using Bayes' Rule, this can be written as

$$P(S = I \mid X = 5) = \frac{P(S = I, X = 5)}{P(X = 5)} = \frac{P(X = 5 \mid S = I) \cdot P(S = I)}{P(X = 5)}$$

Step 1 — Numerator

$$P(X = 5 \mid S = I) = (0.8)^4(0.2) = 0.4096(0.2) = 0.08192$$

Multiply by prior:

$$0.032768$$

Step 2 — Denominator
Casual contribution:

$$(0.4)^4(0.6) = 0.0256(0.6) = 0.01536$$

Multiply by 0.6:

$$0.009216$$

Add intent term:

$$0.009216 + 0.032768 = 0.041984$$

Step 3 — Posterior

$$P(S = I \mid X = 5) = \frac{0.032768}{0.041984} \approx 0.78$$

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Final Interpretation

Originally:

$$P(S = I) = 0.4$$

After observing a long browsing session of 5 pages:

$$P(S = I \mid X = 5) \approx 0.78$$

Long browsing sessions provide strong evidence of intent-driven users.

Case 3 — Continuous, Dependent Exponentials

Shared Infrastructure Failure Model

Motivation

Two cloud services A and B depend on:

- Their own internal systems
- A shared database layer

Let

$$Z_1 \sim \text{Exp}(\lambda_1) \quad \text{(A's internal failure)}$$

$$Z_2 \sim \text{Exp}(\lambda_2) \quad \text{(B's internal failure)}$$

$$Z_0 \sim \text{Exp}(\lambda_0) \quad \text{(shared system failure)}$$

All independent.
Define

$$X = \min(Z_1, Z_0), \qquad Y = \min(Z_2, Z_0)$$

So failures are dependent due to common shock Z_0.
Numerical values: $\lambda_1 = 1, \lambda_2 = 2, \lambda_0 = 1$

Step 1 — Joint Survival Function

$$P(X > x, Y > y)$$

For both services to survive:

$$Z_1 > x, \quad Z_2 > y, \quad Z_0 > \max(x, y)$$

Thus,

$$P(X > x, Y > y) = e^{-\lambda_1 x} e^{-\lambda_2 y} e^{-\lambda_0 \max(x,y)}$$

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Numerical Illustration 1

Compute

$$P(X > 1, Y > 2)$$

Here $\max(1, 2) = 2$.

$$= e^{-1(1)} e^{-2(2)} e^{-1(2)} = e^{-1} e^{-4} e^{-2} = e^{-7}$$

$$e^{-7} \approx 0.00091$$

Very small — surviving 2 hours for B is hard because $\lambda_2 = 2$.

Step 2 — Marginal of X

$$P(X > x) = P(Z_1 > x, Z_0 > x) = e^{-\lambda_1 x} e^{-\lambda_0 x} = e^{-(\lambda_1 + \lambda_0)x}$$

So

$$X \sim \text{Exp}(\lambda_1 + \lambda_0)$$

Numerical Illustration 2

$$P(X > 1) = e^{-(1+1)1} = e^{-2} \approx 0.1353$$

Similarly,

$$P(Y > 1) = e^{-(2+1)1} = e^{-3} \approx 0.0498$$

Step 3 — Conditional Probability

$$P(Y > 1 \mid X > 1) = \frac{P(X > 1, Y > 1)}{P(X > 1)}$$

First compute joint survival:

$$P(X > 1, Y > 1) = e^{-1} e^{-2} e^{-1} = e^{-4} \approx 0.0183$$

Now divide by marginal:

$$P(Y > 1 \mid X > 1) = \frac{e^{-4}}{e^{-2}} = e^{-2} \approx 0.1353$$

Compare to marginal:

$$P(Y > 1) = e^{-3} \approx 0.0498$$

Thus,

$$0.1353 > 0.0498$$

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Survival of A increases survival probability of B.

Interpretation: If A has not failed by 1 hour, shared system likely hasn't failed — good news for B.

Step 4 — Joint Density (Piecewise)

Joint CDF and Joint Density for the Shared Exponential Model

Model Setup

Let

$$X = \min(E_1, E_0), \qquad Y = \min(E_2, E_0),$$

where E_0, E_1, E_2 are independent exponential random variables with rates

$$\lambda_0, \quad \lambda_1, \quad \lambda_2.$$

Step 1: Compute the Joint CDF

By definition,

$$F(x, y) = P(X \le x,\, Y \le y).$$

Using inclusion–exclusion,

$$F(x, y) = 1 - P(X > x) - P(Y > y) + P(X > x, Y > y).$$

Step 2: Compute Each Term

Since

$$X > x \iff E_1 > x \text{ and } E_0 > x,$$

we have

$$P(X > x) = P(E_1 > x)P(E_0 > x) = e^{-\lambda_1 x} e^{-\lambda_0 x} = e^{-(\lambda_1 + \lambda_0)x}.$$

Similarly,

$$P(Y > y) = e^{-(\lambda_2 + \lambda_0)y}.$$

Now,

$$X > x,\ Y > y \iff E_1 > x,\ E_2 > y,\ E_0 > \max(x, y).$$

Thus,

$$P(X > x, Y > y) = e^{-\lambda_1 x} e^{-\lambda_2 y} e^{-\lambda_0 \max(x,y)}.$$

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Step 3: Joint CDF (General Form)

$$F(x,y) = 1 - e^{-(\lambda_1+\lambda_0)x} - e^{-(\lambda_2+\lambda_0)y} + e^{-\lambda_1 x}e^{-\lambda_2 y}e^{-\lambda_0 \max(x,y)}.$$

Step 4: Region-Wise Form

Case 1: $x < y$ Here $\max(x,y) = y$, so

$$F(x,y) = 1 - e^{-(\lambda_1+\lambda_0)x} - e^{-(\lambda_2+\lambda_0)y} + e^{-\lambda_1 x}e^{-(\lambda_2+\lambda_0)y}.$$

Case 2: $y < x$ Here $\max(x,y) = x$, so

$$F(x,y) = 1 - e^{-(\lambda_1+\lambda_0)x} - e^{-(\lambda_2+\lambda_0)y} + e^{-(\lambda_1+\lambda_0)x}e^{-\lambda_2 y}.$$

Step 5: Joint Density

The joint density is obtained by

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y).$$

Region 1: $x < y$ Only the final term depends on both x and y:

$$\frac{\partial}{\partial x} e^{-\lambda_1 x}e^{-(\lambda_2+\lambda_0)y} = -\lambda_1 e^{-\lambda_1 x}e^{-(\lambda_2+\lambda_0)y}.$$

Differentiating again in y gives

$$f(x,y) = \lambda_1(\lambda_2+\lambda_0)e^{-\lambda_1 x}e^{-(\lambda_2+\lambda_0)y}.$$

Region 2: $y < x$ Similarly,

$$f(x,y) = \lambda_2(\lambda_1+\lambda_0)e^{-\lambda_2 y}e^{-(\lambda_1+\lambda_0)x}.$$

Numerical Evaluation

At $(x,y) = (1, 0.5)$:

$$(1+1)(2)e^{-2(1)}e^{-2(0.5)} = 4e^{-2}e^{-1} = 4e^{-3}$$

$$e^{-3} \approx 0.0498$$

$$4 \times 0.0498 = 0.1992$$

Density is larger in this region.

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Step 5 — Probability of Common Shock Failure

They fail together iff

$$Z_0 < Z_1, \quad Z_0 < Z_2$$

For independent exponentials:

$$P(Z_0 \text{ smallest}) = \frac{\lambda_0}{\lambda_0 + \lambda_1 + \lambda_2}$$

Numerical

$$= \frac{1}{1+1+2} = \frac{1}{4} = 0.25$$

25% of outages are shared-infrastructure outages.

Step 6 — Probability A Fails Before B

$$P(X < Y)$$

Using exponential race property:

$$P(Z_1 \text{ smallest}) = \frac{\lambda_1}{\lambda_0 + \lambda_1 + \lambda_2} = \frac{1}{4} = 0.25$$

Thus,

$$P(X < Y) = 0.25$$

Symmetrically,

$$P(Y < X) = \frac{\lambda_2}{\lambda_0 + \lambda_1 + \lambda_2} = \frac{2}{4} = 0.5$$

So B fails first half the time (it has larger λ_2).

Case 4: Bivariate Standard Gaussian with Correlation ρ

Motivation

Suppose:

- X = standardized engagement score of a user
- Y = standardized satisfaction score of the same user

Both are measured in standardized units (mean 0, variance 1 individually). However, engagement and satisfaction may be related:

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- If $\rho > 0$, highly engaged users tend also to report high satisfaction.
- If $\rho < 0$, higher engagement may be associated with lower satisfaction.

We model (X, Y) as a bivariate standard normal distribution with correlation parameter $\rho \in (-1, 1)$.

Joint Density

$$f_\rho(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right).$$

Parameter ρ determines how strongly engagement and satisfaction move together.

Marginal Density of X (Detailed Derivation)

$$f_X(x) = \int_{-\infty}^{\infty} f_\rho(x, y)\, dy.$$

Substitute the joint density:

$$f_X(x) = \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right) dy.$$

Complete the square in y:

$$x^2 - 2\rho xy + y^2 = (y - \rho x)^2 + x^2(1 - \rho^2).$$

So the exponent becomes

$$-\frac{(y-\rho x)^2}{2(1-\rho^2)} - \frac{x^2}{2}.$$

Thus

$$f_X(x) = \frac{e^{-x^2/2}}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left(-\frac{(y-\rho x)^2}{2(1-\rho^2)}\right) dy.$$

The integral equals

$$\sqrt{2\pi(1-\rho^2)}.$$

Therefore,

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

So

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$$X \sim N(0,1).$$

Similarly,

$$Y \sim N(0,1).$$

Conditional Density of Y Given $X = x$

$$f_{Y|X}(y|x) = \frac{f_\rho(x,y)}{f_X(x)}.$$

Substitute:

$$f_{Y|X}(y|x) = \frac{\frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{(y-\rho x)^2}{2(1-\rho^2)} - \frac{x^2}{2}\right)}{\frac{1}{\sqrt{2\pi}} e^{-x^2/2}}.$$

The $e^{-x^2/2}$ cancels:

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi(1-\rho^2)}} \exp\left(-\frac{(y-\rho x)^2}{2(1-\rho^2)}\right).$$

Thus,

$$Y|X = x \sim N(\rho x,\, 1-\rho^2).$$

Interpretation

- If $\rho > 0$, observing high engagement shifts satisfaction upward.
- If $\rho < 0$, observing high engagement shifts satisfaction downward.
- If $\rho = 0$, satisfaction does not depend on engagement.

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